1 ElGamal and Elliptic Curve Cryptosystems ▸

▸ **Likely this year:** New this year: an ElGamal encrypt/decrypt with a small prime, "find all points on $E_p(a,b)$", one point addition or doubling, and "why must r be fresh / why ECC has smaller keys".

1.1 ElGamal cryptosystem ▸

▸ **Likely this year:** "Explain ElGamal key generation, encryption and decryption; prove decryption is correct; encrypt P with p, e_1, d, r given" (4–5 marks).

RSA rests on factoring. ElGamal rests on a different hard problem, the **discrete logarithm problem** (DLP, §**??**). Let p be a large prime and e_1 a **primitive root** of p, i.e. an element whose powers $e_1^1, e_1^2, \dots, e_1^{p-1}$ run through all of $\mathbb{Z}_p^* = \{1, \dots, p-1\}$ (§**??**). Computing $e_2 = e_1^d \bmod p$ is easy by square-and-multiply, but given e_1, e_2 and p, finding $d = \log_{e_1} e_2 \bmod p$ is infeasible for large p. So $d \mapsto e_1^d$ is a one-way function, and Bob's private key is the exponent d hidden inside the public e_2.

The idea of encryption is to multiply P by a **one-time mask** e_2^r that only Bob can rebuild. Alice picks a fresh random r and sends two numbers: $C_1 = e_1^r$, a "hint" that reveals nothing about r (DLP again), and $C_2 = P \cdot e_2^r$, the masked plaintext. Bob rebuilds the mask from the hint with his secret: $C_1^d = e_1^{rd} = (e_1^d)^r = e_2^r$. Eve would need either d or r, and both are discrete logarithms. This is the Diffie–Hellman idea turned into a cipher: $e_2^r = e_1^{dr}$ is a shared value that Alice computes from (r, e_2) and Bob from (d, C_1).

★ MUST KNOW : ElGamal

Key generation (Bob): choose a large prime p and a primitive root e_1 of p; choose d with $1 \le d \le p-2$; compute $e_2 = e_1^d \bmod p$. Public key (e_1, e_2, p), private key d.
Encryption (Alice): choose a **fresh** random $r \in \mathbb{Z}_p^*$; $C_1 = e_1^r \bmod p$, $C_2 = (P \times e_2^r) \bmod p$. Send (C_1, C_2).
Decryption (Bob): $P = \left[C_2 \times (C_1^d)^{-1}\right] \bmod p$, and $(C_1^d)^{-1} = C_1^{p-1-d} \bmod p$.

Proof of correctness. $C_2 \times (C_1^d)^{-1} = (P \cdot e_2^r)(e_1^{rd})^{-1} = P \cdot e_2^r (e_2^r)^{-1} = P \pmod p$, because $e_1^{rd} = (e_1^d)^r = e_2^r$. The inverse exists since $C_1^d \in \mathbb{Z}_p^*$.

The inverse shortcut. By Fermat, $C_1^{p-1} \equiv 1$, so $C_1^d \cdot C_1^{p-1-d} \equiv 1$, i.e. $(C_1^d)^{-1} = C_1^{p-1-d}$. One exponentiation replaces an exponentiation followed by an extended-Euclid inverse.

Properties. ElGamal is **probabilistic**: the same P encrypted twice gives different ciphertexts because r changes, so Eve cannot test guesses by re-encrypting them (unlike textbook RSA). The price is **ciphertext expansion**: the ciphertext is the pair (C_1, C_2), each as long as p, so it is **twice the size** of the plaintext. Like RSA, it is slow and is used for short data such as session keys.

WORKED EXAMPLE : the slide example

Q. Bob chooses $p = 11$, $e_1 = 2$, $d = 3$. Find his public key. Alice encrypts $P = 7$ with $r = 4$. Find the ciphertext and show Bob's decryption.
Keys. 2 is a primitive root of 11 (its powers are $2, 4, 8, 5, 10, 9, 7, 3, 6, 1$). $e_2 = 2^3 \bmod 11 = 8$. Public key $(2, 8, 11)$, private key 3.
Encrypt. $C_1 = 2^4 \bmod 11 = 16 \bmod 11 = 5$. $e_2^r = 8^4 = 4096 = 372 \cdot 11 + 4$, so $C_2 = 7 \cdot 4 = 28 \bmod 11 = 6$. Ciphertext $(\mathbf{5}, \mathbf{6})$.
Decrypt. $C_1^d = 5^3 = 125 = 11 \cdot 11 + 4 \equiv 4$; $4^{-1} \bmod 11 = 3$ $(4 \cdot 3 = 12)$. $P = 6 \cdot 3 = 18 \bmod 11 = \mathbf{7}$ ✓.
Shortcut. $(C_1^d)^{-1} = 5^{11-1-3} = 5^7$: $5^2 \equiv 3$, $5^4 \equiv 9$, $5^7 = 9 \cdot 3 \cdot 5 = 135 = 12 \cdot 11 + 3 \equiv 3$ ✓.

WORKED EXAMPLE : new exam-style example

Q. In ElGamal, $p = 23$, $e_1 = 5$, Bob's private key $d = 6$. (a) Find e_2. (b) Encrypt $P = 20$ with $r = 3$. (c) Decrypt, and verify with the shortcut.
(a) $5^2 = 25 \equiv 2$, $5^4 \equiv 4$, $e_2 = 5^6 = 5^4 \cdot 5^2 \equiv 4 \cdot 2 = \mathbf{8}$. Public key $(5, 8, 23)$. (5 is a primitive root of 23: its order is 22.)
(b) $C_1 = 5^3 = 125 = 5 \cdot 23 + 10 \Rightarrow C_1 = 10$. $e_2^r = 8^3 = 512 = 22 \cdot 23 + 6 \equiv 6$. $C_2 = 20 \cdot 6 = 120 = 5 \cdot 23 + 5 \Rightarrow C_2 = 5$. Ciphertext $(\mathbf{10}, \mathbf{5})$.
(c) $C_1^d = 10^6$: $10^2 = 100 \equiv 8$, $10^4 \equiv 64 \equiv 18$, $10^6 = 18 \cdot 8 = 144 \equiv 6$ (this is e_2^r again, as the proof says). $6^{-1} \bmod 23 = 4$ $(6 \cdot 4 = 24)$. $P = 5 \cdot 4 = \mathbf{20}$ ✓.
Shortcut: $10^{23-1-6} = 10^{16}$: $10^8 = 18^2 = 324 = 14 \cdot 23 + 2 \equiv 2$, $10^{16} \equiv 4$ ✓ (the same inverse).

1.2 Security of ElGamal ▸

▸ **Likely this year:** "Why must Alice use a fresh r for every message?" with the algebra.

Low-modulus attack. If p is small, Eve solves the discrete logarithm $d = \log_{e_1} e_2$ once (by search or a sub-

exponential algorithm), stores d, and decrypts every message sent to Bob for as long as he keeps the key. With the slide key she tries $2^1 = 2$, $2^2 = 4$, $2^3 = 8 = e_2$ and has $d = 3$. Hence p must be at least 1024 bits (about 309 digits).

Known-plaintext attack when r is reused. The mask e_2^r depends only on r, so two messages encrypted with the same r share the same mask. If Eve learns one plaintext P and sees both ciphertexts $C_2 = P \cdot e_2^r$ and $C_2' = P' \cdot e_2^r$, she removes the mask without knowing d or r:

$$e_2^r = C_2 \times P^{-1} \bmod p, \qquad P' = C_2' \times (e_2^r)^{-1} \bmod p.$$

In the example above ($p = 23$, $e_2 = 8$, $r = 3$), suppose Alice also sends $P' = 17$ with the same r: $C_2' = 17 \cdot 6 = 102 \equiv 10$. Eve, knowing $P = 20$ and $C_2 = 5$: $20^{-1} \bmod 23 = 15$, so $e_2^r = 5 \cdot 15 = 75 \equiv 6$; then $P' = 10 \cdot 6^{-1} = 10 \cdot 4 = 40 \equiv 17$ ✓. The repeated C_1 even tells Eve that r was reused. Alice must use a fresh random r for every message.

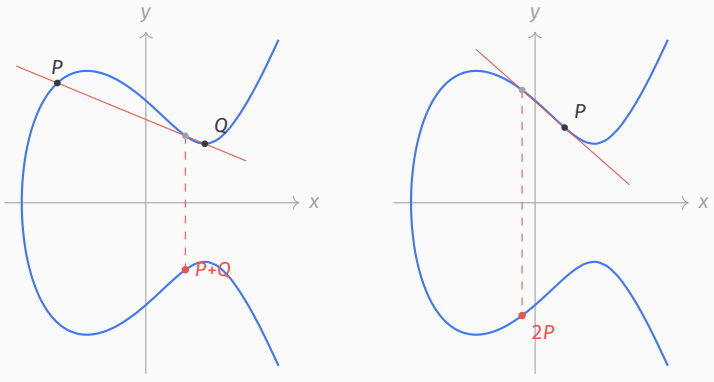
	RSA	ElGamal
Hard problem	factoring $n = pq$	discrete logarithm mod p
Public / private key	(e, n) / d	(e_1, e_2, p) / d
Encryption	deterministic, $C = P^e$	probabilistic, fresh r each time
Ciphertext size	same as plaintext	two numbers: twice the plaintext
Key generation	two primes, $d = e^{-1} \bmod \phi(n)$	one prime, $e_2 = e_1^d$
Main misuse	small e, no padding, shared n	reused r, small p

1.3 Elliptic curves over the real numbers

RSA and ElGamal are secure, but their security costs large keys (1024 bits and more). **Elliptic curve cryptography** (ECC) gets the same security with much smaller keys by replacing the group $\mathbb{Z}_p^*$ with the group of points on an elliptic curve, where the discrete logarithm is harder.

An **elliptic curve** is the set of points (x, y) satisfying the cubic $y^2 = x^3 + ax + b$. It is symmetric about the x-axis, since $(x, -y)$ satisfies the equation whenever (x, y) does. The curve must be **non-singular**, meaning $x^3 + ax + b$ has three distinct roots, which is equivalent to $4a^3 + 27b^2 \neq 0$. A singular curve has a cusp or a self-crossing where the tangent is undefined, and the group law below breaks there; for example $a = -3, b = 2$ gives $4(-27) + 27 \cdot 4 = 0$ and $x^3 - 3x + 2 = (x - 1)^2(x + 2)$ has a double root. The slide's curve $y^2 = x^3 - 3x + 3$ has $4a^3 + 27b^2 = -108 + 243 = 135 \neq 0$ and one real root, so it is a single connected curve.

Adding points (chord-and-tangent rule). To add $P \neq Q$, draw the line through P and Q; a line meets a cubic in three points, so it cuts the curve at a third point, and $R = P + Q$ is the **reflection** of that third point in the x-axis. To double P, use the **tangent** at P (the line through P and a point infinitely close to it), take its second intersection and reflect it. If the line is vertical ($Q = -P$, the reflection of P), there is no third point; we say it meets the curve at the **point at infinity** O, which is the identity: $P + (-P) = O$, $P + O = P$. With this rule the points form an abelian group (§**??**).



addition: chord — doubling: tangent

Deriving the formulas. Let the line through $P = (x_1, y_1)$ have slope s: $y = s(x - x_1) + y_1$. Substituting into $y^2 = x^3 + ax + b$ gives a cubic in x whose expanded form starts $x^3 - s^2x^2 + \cdots = 0$. Its three roots are the x-coordinates of the three intersection points, and the sum of the roots of a monic cubic is minus the x^2 coefficient, so $x_1 + x_2 + x_3 = s^2$, giving $x_3 = s^2 - x_1 - x_2$. The third point is $(x_3,\ s(x_3 - x_1) + y_1)$; reflecting it gives $y_3 = s(x_1 - x_3) - y_1$. For addition, s is the chord slope $(y_2 - y_1)/(x_2 - x_1)$. For doubling, s is the tangent slope: differentiating $y^2 = x^3 + ax + b$ implicitly gives $2y\,y' = 3x^2 + a$, so $s = (3x_1^2 + a)/(2y_1)$, and $x_2 = x_1$. The constant b vanishes on differentiation, which is why the doubling slope contains a and never b.

1.4 Elliptic curves over GF(p) ▸

▸ **Likely this year:** Find all the points on $E_{13}(1,1)$ / $E_{11}(1,6)$" and find the inverse of a point" (3–4 marks).

Cryptography cannot use real numbers (rounding errors, unbounded size), so the same equation is taken modulo a prime $p > 3$. The curve $E_p(a,b)$ is the set of all pairs $(x,y) \in \mathbb{Z}_p \times \mathbb{Z}_p$ with

$$y^2 \equiv x^3 + ax + b \pmod{p}, \qquad a,b \in \mathbb{Z}_p, \quad 4a^3 + 27b^2 \not\equiv 0 \pmod{p},$$

together with the point at infinity O. The curve" is now a finite scatter of points, but the addition formulas derived above still define a group, with every division done as multiplication by an inverse modulo p.

Finding the points. For each $x = 0, 1, \ldots, p-1$, compute $w = (x^3 + ax + b) \bmod p$, which must equal y^2. If $w = 0$ there is one point $(x, 0)$. If w is a nonzero **quadratic residue** (a perfect square mod p), there are two points $(x, \sqrt{w})$ and $(x, p - \sqrt{w})$. If w is not a square, no point has that x. Make the table of squares y^2 mod p first; since $(p-y)^2 \equiv y^2$, only $y = 0, \ldots, (p-1)/2$ are needed.

Negatives. The inverse of $P = (x,y)$ is $-P = (x,-y) = (x, p-y)$: the same x, only y flipped (the reflection in the x-axis). A point with $y = 0$ is its own inverse. Points therefore come in vertical pairs $(x,y), (x,p-y)$.

Hasse's theorem. The number of points $\#E$ (including O) satisfies $p + 1 - 2\sqrt{p} \le \#E \le p + 1 + 2\sqrt{p}$: about half the x values give a square and those give two points each, so #E is close to $p + 1$. It is a quick check on a point count.

WORKED EXAMPLE : Forouzan/slide, all points of $E_{13}(1,1)$

Q. Find all points on $E_{13}(1,1)$: $y^2 = x^3 + x + 1 \pmod{13}$. List the inverse pairs.
Valid curve: $4 + 27 = 31 \equiv 5 \neq 0 \pmod{13}$. **Squares mod 13**: $y = 0..6$ give $y^2 = 0, 1, 4, 9, 3, 12, 10$, so the nonzero squares are $\{1, 3, 4, 9, 10, 12\}$ with roots $1 \to \pm 1$, $3 \to \pm 4$, $4 \to \pm 2$, $9 \to \pm 3$, $10 \to \pm 6$, $12 \to \pm 5$.

x	$x^3 + x + 1$	w	square?	points
0	1	1	yes	(0, 1), (0, 12)
1	3	3	yes	(1, 4), (1, 9)
2	11	11	no	–
3	31	5	no	–
4	69	4	yes	(4, 2), (4, 11)
5	131	1	yes	(5, 1), (5, 12)
6	223	2	no	–
7	351	0	$y = 0$	(7, 0)
8	521	1	yes	(8, 1), (8, 12)
9	739	11	no	–
10	1011	10	yes	(10, 6), (10, 7)
11	1343	4	yes	(11, 2), (11, 11)
12	1741	12	yes	(12, 5), (12, 8)

17 points + O = 18. Each row's two points are inverses (e.g. $4 + 9 = 13$, so $(1,9) = -(1,4)$); $(7,0)$ is its own inverse. $x = 2, 3, 6, 9$ give non-squares. Hasse: $14 \pm 2\sqrt{13} = [6.8,\ 21.2]$ contains 18 ✓.

WORKED EXAMPLE : new, all points of $E_{11}(1,6)$

Q. Find all points on $y^2 = x^3 + x + 6 \pmod{11}$ and verify Hasse's bound.
Valid: $4 + 27 \cdot 36 = 976 = 88 \cdot 11 + 8 \neq 0$. **Squares mod 11**: $y = 0..5$ give $0, 1, 4, 9, 5, 3$; nonzero squares $\{1, 3, 4, 5, 9\}$ with roots $1 \to \pm 1$, $3 \to \pm 5$, $4 \to \pm 2$, $5 \to \pm 4$, $9 \to \pm 3$.

x	x^3+x+6	w	sq?	points	x	x^3+x+6	w	sq?	points
0	6	6	no	–	6	228	8	no	–
1	8	8	no	–	7	356	4	yes	(7, 2), (7, 9)
2	16	5	yes	(2, 4), (2, 7)	8	526	9	yes	(8, 3), (8, 8)
3	36	3	yes	(3, 5), (3, 6)	9	744	7	no	–
4	74	8	no	–	10	1016	4	yes	(10, 2), (10, 9)
5	136	4	yes	(5, 2), (5, 9)					

12 points + O = 13. Hasse: $12 \pm 2\sqrt{11} = [5.4,\ 18.6]$ contains 13 ✓. Since 13 is prime, every point except O generates the whole group (Lagrange).

1.5 Point addition and doubling over GF(p) ▸

▸ **Likely this year:** "Add P and Q / find $2P$ on $E_p(a,b)$" (2–3 marks), always with modular inverses.

★ MUST KNOW : EC group law over GF(p)

For $P = (x_1, y_1)$, $Q = (x_2, y_2)$, $R = P + Q = (x_3, y_3)$ with

$$x_3 = s^2 - x_1 - x_2 \bmod p, \qquad y_3 = s(x_1 - x_3) - y_1 \bmod p,$$

$$s = \frac{y_2 - y_1}{x_2 - x_1} \bmod p \ \ (P \neq Q, \text{ addition}), \qquad s = \frac{3x_1^2 + a}{2y_1} \bmod p \ \ (P = Q, \text{ doubling}).$$

Division means multiplying by the inverse mod p. Special cases: $P + O = P$; if $x_1 = x_2$ and $y_2 = -y_1$ then $P + Q = O$; doubling a point with $y_1 = 0$ gives O.

WORKED EXAMPLE : one addition and two doublings

Q1 (Forouzan style). On $E_{13}(1, 1)$, add $P = (4, 2)$ and $Q = (10, 6)$.
$s = (6 - 2)(10 - 4)^{-1} = 4 \cdot 6^{-1} = 4 \cdot 11 = 44 \equiv 5 \ \ (6 \cdot 11 = 66 = 5 \cdot 13 + 1)$.
$x_3 = 25 - 4 - 10 = 11$; $y_3 = 5(4 - 11) - 2 = -37 \equiv -37 + 39 = 2$. $\mathbf{P + Q = (11, 2)}$, which is in the point list ✓.
Q2. On $E_{13}(1, 1)$, find $2P$ for $P = (1, 4)$.
$s = (3 \cdot 1^2 + 1)(2 \cdot 4)^{-1} = 4 \cdot 8^{-1} = 4 \cdot 5 = 20 \equiv 7 \ \ (8 \cdot 5 = 40 = 3 \cdot 13 + 1)$.
$x_3 = 49 - 1 - 1 = 47 \equiv 8$; $y_3 = 7(1 - 8) - 4 = -53 \equiv -53 + 65 = 12$. $\mathbf{2P = (8, 12)}$ ✓.
Q3 (slide). On $E_{17}(2, 2)$: $y^2 = x^3 + 2x + 2$, find $2P$ for $P = (5, 1)$.
$s = (3 \cdot 5^2 + 2)(2 \cdot 1)^{-1} = 77 \cdot 2^{-1} \equiv 9 \cdot 9 = 81 \equiv 13 \ \ (77 = 4 \cdot 17 + 9, 2^{-1} = 9)$.
$x_3 = 169 - 5 - 5 = 159 = 9 \cdot 17 + 6 \equiv 6$; $y_3 = 13(5 - 6) - 1 = -14 \equiv 3$. $\mathbf{2P = (6, 3)}$.
Check: $3^2 = 9$ and $6^3 + 2 \cdot 6 + 2 = 230 = 13 \cdot 17 + 9$ ✓. (This curve has 19 points and $(5, 1)$ generates all of them.)

Trap: point arithmetic slips

The doubling slope uses a (from $3x^2 + a$), never b. Compute s as an integer mod p with a modular inverse, never as a fraction or decimal. $-P = (x, p - y)$ flips only y. Use the addition formula only for $P \neq Q$ (it would divide by $x_2 - x_1 = 0$). Reduce negative results by adding multiples of p, and check the answer lies on the curve.

1.6 Scalar multiplication and the ECDLP ▸

▸ **Likely this year:** "Compute kP for small k and "state the ECDLP / Hasse's theorem.

Multiplying a point by an integer k means adding it to itself: $kP = P + P + \cdots + P$ (k times). It is computed by **double-and-add**, the analogue of square-and-multiply: scan the bits of k, double at every bit and add P at every 1-bit, so kP costs about $\log_2 k$ doublings. The slides' two examples on $E_{13}(1, 1)$, recomputed:

- $4 \cdot (1, 4) = 2 \cdot (2 \cdot (1, 4)) = 2 \cdot (8, 12)$: $s = (3 \cdot 64 + 1) \cdot 24^{-1} \equiv 11 \cdot 11^{-1} = 1$, $x_3 = 1 - 16 \equiv 11$, $y_3 = 1 \cdot (8 - 11) - 12 = -15 \equiv 11$, so $4 \cdot (1, 4) = \mathbf{(11, 11)}$ ✓.
- $3 \cdot (8, 1) = 2 \cdot (8, 1) + (8, 1)$: doubling gives $s = 193 \cdot 2^{-1} \equiv 11 \cdot 7 \equiv 12$, $(11, 2)$; then $(11, 2) + (8, 1)$: $s = (-1)(-3)^{-1} \equiv 12 \cdot 10^{-1} = 12 \cdot 4 \equiv 9$, $x_3 = 81 - 19 \equiv 10$, $y_3 = 9(11 - 10) - 2 = 7$, so $3 \cdot (8, 1) = \mathbf{(10, 7)}$ ✓.

$(1, 4)$ has order 18, so it generates all of $E_{13}(1, 1)$; its multiples (useful for checking the examples below) are:

k	1	2	3	4	5	6	7	8	9
$k(1, 4)$	(1,4)	(8,12)	(0,12)	(11,11)	(5,1)	(10,6)	(12,8)	(4,2)	(7,0)
k	17	16	15	14	13	12	11	10	18
$k(1, 4)$	(1,9)	(8,1)	(0,1)	(11,2)	(5,12)	(10,7)	(12,5)	(4,11)	O

The second row is the negative of the first ($(18 - k)P = -kP$), and $9P = (7, 0)$ has $y = 0$, so $18P = 2 \cdot 9P = O$.

Elliptic curve discrete logarithm problem (ECDLP). Given a point P (ideally a generator) and $T = dP$, find d with $1 \le d \le \#E$. Computing T from d takes about $\log_2 d$ doublings, but no shortcut is known for the reverse: on a well-chosen curve the best attack is a generic one such as **Pollard's rho**, which needs about $\sqrt{\#E}$ steps. On the toy curve the table answers it at once ($T = (4, 2) \Rightarrow d = 8$); for a 160-bit curve, $\sqrt{\#E} \approx 2^{80}$ steps is infeasible.

1.7 ECC simulating ElGamal ▸

▸ **Likely this year:** "Explain ECC encryption/decryption; verify the given example" (the slide's $E_{67}(2, 3)$ example).

ElGamal works in any group where the discrete log is hard, so it carries over by a dictionary: multiplication mod p becomes point addition, exponentiation becomes scalar multiplication, and the inverse becomes the negative point. The plaintext must be encoded as a point P on the curve.

★ MUST KNOW : EC ElGamal

Keys (Bob): choose $E_p(a,b)$, a point e_1 on it, and an integer d; compute $e_2 = d \times e_1$. Public $(E_p(a,b), e_1, e_2)$, private d.
Encrypt (Alice): fresh random r; $C_1 = r \times e_1$, $C_2 = P + r \times e_2$.
Decrypt (Bob): $P = C_2 - (d \times C_1)$, where subtracting means adding the negative point.
Proof: $C_2 - dC_1 = P + r(de_1) - d(re_1) = P + (rd)e_1 - (dr)e_1 = P + O = P$.

WORKED EXAMPLE : the slide example on $E_{67}(2,3)$, every step

Q. Bob selects $E_{67}(2,3)$, $e_1 = (2,22)$, $d = 4$. Alice sends $P = (24,26)$ with $r = 2$. Find e_2, C_1, C_2, and verify decryption.

Step	Slope s (mod 67)	x_3	y_3	Result
$2e_1 = 2(2,22)$	$14 \cdot 44^{-1} = 14 \cdot 32 \equiv 46$	$2112 \equiv 35$	$-1540 \equiv 1$	$(35,1)$
$4e_1 = 2(35,1)$	$3677 \cdot 2^{-1} \equiv 59 \cdot 34 \equiv 63$	$3899 \equiv 13$	$1385 \equiv 45$	$(13,45)$
$2e_2 = 2(13,45)$	$509 \cdot 90^{-1} \equiv 40 \cdot 35 \equiv 60$	$3574 \equiv 23$	$-645 \equiv 25$	$(23,25)$
$P + 2e_2$	$(-1)(-1)^{-1} = 1$	$-46 \equiv 21$	$-23 \equiv 44$	$(21,44)$
$C_2 + (23,42)$	$(-2) \cdot 2^{-1} \equiv 66$	$4312 \equiv 24$	$-242 \equiv 26$	$(24,26)$

Keys: $e_2 = 4e_1 = (13,45)$. **Encrypt**: $C_1 = 2e_1 = (35,1)$, $re_2 = (23,25)$, $C_2 = (24,26) + (23,25) = (21,44)$.
Decrypt: $d \times C_1 = 4 \cdot (35,1) = 8e_1 = 2e_2 = (23,25)$ (two doublings, rows 2–3). Its negative is $(23, 67 - 25) = (23,42)$, and $(21,44) + (23,42) = (\mathbf{24,26}) = P$ ✓.
Inverses used: $44 \cdot 32 = 1408 = 21 \cdot 67 + 1$; $2 \cdot 34 = 68$; $23 \cdot 35 = 805 = 12 \cdot 67 + 1$ ($90 \equiv 23$).

WORKED EXAMPLE : new exam-style example on $E_{13}(1,1)$

Q. Bob uses $E_{13}(1,1)$ with $e_1 = (1,4)$ and private key $d = 3$. Alice encrypts the point $P = (4,2)$ with $r = 2$. Find Bob's public key, the ciphertext, and verify decryption.

Step	Slope s (mod 13)	x_3	y_3	Result
$2e_1 = 2(1,4)$	$4 \cdot 8^{-1} = 4 \cdot 5 \equiv 7$	$47 \equiv 8$	$-53 \equiv 12$	$(8,12)$
$e_2 = (8,12) + (1,4)$	$(-8)(-7)^{-1} = 5 \cdot 6^{-1} = 5 \cdot 11 \equiv 3$	0	12	$(0,12)$
$2e_2 = 2(0,12)$	$1 \cdot 24^{-1} = 11^{-1} = 6$	$36 \equiv 10$	$-72 \equiv 6$	$(10,6)$
$C_2 = (4,2) + (10,6)$	$4 \cdot 6^{-1} \equiv 5$	11	$-37 \equiv 2$	$(11,2)$
$2C_1 = 2(8,12)$	$193 \cdot 24^{-1} \equiv 11 \cdot 6 \equiv 1$	$-15 \equiv 11$	$-15 \equiv 11$	$(11,11)$
$3C_1 = (11,11) + (8,12)$	$1 \cdot (-3)^{-1} = 10^{-1} = 4$	$-3 \equiv 10$	$-7 \equiv 6$	$(10,6)$
$C_2 + (10,7)$	$5 \cdot (-1)^{-1} = 5 \cdot 12 \equiv 8$	$43 \equiv 4$	$54 \equiv 2$	$(4,2)$

Public key $(E_{13}(1,1),(1,4),(0,12))$. **Ciphertext** $C_1 = 2e_1 = (\mathbf{8,12})$, $C_2 = (\mathbf{11,2})$.
Decrypt: $d \times C_1 = 3 \cdot (8,12) = (10,6)$, the same point as $r \times e_2$; negative $(10,7)$; $(11,2) + (10,7) = (\mathbf{4,2})$ ✓. In multiples of e_1 (table above): $e_2 = 3e_1$, $C_1 = 2e_1$, $P = 8e_1$, $C_2 = (8+6)e_1 = 14e_1$, and $14 - 6 = 8$ ✓.

1.8 Security of ECC and key size ▸

▸ **Likely this year:** "Why does ECC need smaller keys than RSA for the same security? (2 marks).

To decrypt, Eve needs r (then $P = C_2 - r \times e_2$) or d (then $P = C_2 - d \times C_1$). Finding r from $C_1 = r \times e_1$, or d from $e_2 = d \times e_1$, is the ECDLP, which for large p (or large n in GF(2^n)) is infeasible by Pollard's rho. As in ElGamal, r must be fresh for every message.

Why ECC keys are smaller. For factoring and for the discrete log in $\mathbb{Z}_p^*$ there are **sub-exponential** algorithms (the number field sieve, index calculus) that exploit the arithmetic structure of integers, so RSA and ElGamal need 1024-bit and larger moduli to stay ahead. On a well-chosen elliptic curve no such structure is known, and only generic attacks costing about $\sqrt{\#E}$ work, i.e. $2^{k/2}$ for a k-bit curve. A 160-bit curve therefore costs about 2^{80} steps, the same effort as factoring a 1024-bit RSA modulus: ECC over GF(2^n) or GF(p) with a 160-bit modulus gives the same security as RSA with a 1024-bit modulus (and 256-bit ECC ≈ 3072-bit RSA). Smaller keys mean less computation, bandwidth and storage, which suits smart cards and mobile devices.

Check your understanding

1. **Differentiate RSA and ElGamal.** RSA rests on factoring $n = pq$, is deterministic, and its ciphertext is one number the size of n. ElGamal rests on the discrete logarithm modulo a prime, is probabilistic because of the fresh r, and its ciphertext is a pair (C_1, C_2), twice the plaintext size.
2. **Why must r be fresh for every ElGamal message?** The mask e_2^r depends only on r; if two messages share r,

one known plaintext P gives $e_2^r = C_2 P^{-1}$, and then every other message encrypted with that r is $P' = C_2'(e_2^r)^{-1}$.

3. **Why is** $(C_1^d)^{-1} = C_1^{p-1-d}$**?** $C_1^d \cdot C_1^{p-1-d} = C_1^{p-1} \equiv 1 \pmod p$ by Fermat's little theorem.
4. **ElGamal:** $p = 11$**,** $e_1 = 2$**,** $d = 3$**,** $r = 4$**,** $P = 7$**.** $(C_1, C_2) = (5, 6)$; decryption $6 \cdot (5^3)^{-1} = 6 \cdot 3 \equiv 7$.
5. **Why must** $4a^3 + 27b^2 \neq 0$**?** Otherwise the cubic has a repeated root, the curve is singular (a cusp or crossing), the tangent there is undefined, and the points do not form a group.
6. **What is** $-(5, 1)$ **on** $E_{13}(1, 1)$ **and what is** $(7, 0) + (7, 0)$**?** $-(5, 1) = (5, 12)$ (only y flips); $(7, 0) + (7, 0) = O$ because the tangent at a point with $y = 0$ is vertical.
7. **Why does the doubling slope contain** a **but not** b**?** It is the derivative dy/dx from $2y\,y' = 3x^2 + a$, and the constant b differentiates to 0.
8. **Is** $(3, 5)$ **on** $E_{11}(1, 6)$**? Find** $2 \cdot (2, 7)$**.** Yes: $27 + 3 + 6 = 36 \equiv 3 \equiv 5^2$. For $2(2, 7)$: $s = 13 \cdot 14^{-1} \equiv 2 \cdot 3^{-1} = 2 \cdot 4 = 8$, $x_3 = 64 - 4 = 60 \equiv 5$, $y_3 = 8(2 - 5) - 7 = -31 \equiv 2$, so $(5, 2)$.
9. **Why does 160-bit ECC match 1024-bit RSA?** Factoring has sub-exponential algorithms, while the ECDLP on a good curve only has generic $\sqrt{n}$-time attacks, so a 160-bit curve already needs about 2^{80} work.
10. **Why does the ElGamal ciphertext double in size?** Two values C_1 and C_2, each in $\mathbb{Z}_p$, are sent for one plaintext block in $\mathbb{Z}_p$.

Corrections to the notes / slides

- ElGamal block diagram: the encryption box labels both lines C_1; the second line, $(e_2^r \cdot P) \bmod p$, is C_2.
- $E_{67}(2, 3)$ example, step 8: "He uses $2 \times C_1$" should read $d \times C_1 = 4 \times C_1 = (23, 25)$. With $C_1 = 2e_1$, $2 \times C_1 = 4e_1 = e_2 = (13, 45)$, not $(23, 25)$ (recomputed). The same misprint is in Forouzan.
- ElGamal security slide: 1024 bits is about **309** decimal digits (as on the RSA slide), not 300.